1. Descriptive Statistics

Frequency notation
f_i = absolute frequency
$p_i = f_i/n$ = relative frequency
F_i = cumulative frequency
$n = \Sigma f_i$ = sample size

Central Tendency Measures

Mean: $\bar{x} = (1/n) \Sigma x_i$
Grouped data: $\bar{x} = \Sigma x^*_k p_k$
Approximation using midpoints: $\bar{x} \approx \Sigma m_k p_k$
Median (odd n): $Me = x_{(n+1)/2}$
Median (even n): $Me = (x_{(n/2)} + x_{(n/2)+1}) / 2$
Mode: value with highest frequency

Dispersion Measures

Range: $R = \max(x_i) - \min(x_i)$
Variance (sample): $s^2 = (1/(n-1)) \Sigma (x_i - \bar{x})^2$
Standard deviation: $s = \sqrt{s^2}$
Coefficient of variation: $CV = s / \bar{x}$
Interquartile range: $IQR = Q3 - Q1$

Correlation

Covariance: $s_{xy} = (1/(n-1))\ \Sigma\ (x_i - x$■$)(y_i - $■$)$
Correlation coefficient: $r = s_{xy} / (s_x\ s_y)$

2. Probability and Random Variables

Probability: $P(A \cup B)=P(A)+P(B)-P(A \cap B)$
Conditional probability: $P(A|B)=P(A \cap B)/P(B)$
Independent events: $P(A \cap B)=P(A)P(B)$
Expected value: $E[X]=\Sigma x_i P(X=x_i)$
Variance: $Var(X)=E[(X-E[X])^2]=E[X^2]-(E[X])^2$

3. Common Distributions

Bernoulli(p): $P(X=x)=p^x(1-p)^{1-x}$, $E[X]=p$, $Var(X)=p(1-p)$
Normal(μ,σ^2): $f(x)=1/(\sqrt{(2\pi\sigma^2)})\ e^{-(x-\mu)^2/(2\sigma^2)}$, $E[X]=\mu$, $Var(X)=\sigma^2$
Standardization: $Z=(X-\mu)/\sigma$

4. Sampling Distributions

Sample mean: $E[x\blacksquare]=\mu$, $Var(x\blacksquare)=\sigma^2/n$, $SE=\sigma/\sqrt{n}$
Sample proportion: $E[p\blacksquare]=p$, $Var(p\blacksquare)=p(1-p)/n$, $SE=\sqrt{(p(1-p)/n)}$

5. Point Estimation

Unbiasedness: E[θ■]=θ
Bias: Bias(θ■)=E[θ■]−θ
Asymptotic unbiasedness: lim(n→∞)Bias(θ■)=0
Efficiency: smaller Var(θ■) ⇒ more efficient

6. Estimators and Standard Errors

Population mean: θ■=x■, $SE=\sigma/\sqrt{n}$ or $s/\sqrt{n}$
Population proportion: θ■=p■, $SE=\sqrt{(p(1-p)/n)}$ or $\sqrt{(p$■$(1-p$■$)/n)}$
Difference of means (independent): $SE=\sqrt{(s_x^2/n_x + s_y^2/n_y)}$
Paired difference: $SE=s_D/\sqrt{n}$
Difference of proportions: $SE=\sqrt{(p$■$(1-p$■$)/n$■ $+ p$■$(1-p$■$)/n$■$)}$

7. Normal Distribution Functions in R

pnorm(q, mean, sd) → cumulative probability $P(X \le q)$
1 - pnorm(q, mean, sd) → $P(X > q)$
qnorm(p, mean, sd) → quantile (percentile)

8. UBStats Functions

distr.table.x() – frequency tables
distr.plot.x() – histograms/bar plots
distr.summary.x(stats='central') – mode, median, mean
distr.summary.x(stats='dispersion') – range, IQR, sd, var
distr.table.xy() – cross-tabulations
distr.plot.xy() – bar plots for two variables